

CPSC 416 Distributed Systems

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Project 3 Overview



What you have

Client and server skeleton

Client and server communication protocols

KVStore and AMOApplication from Project 1 & 2



What you need (1)

View server that manages a list of servers in the cluster

- Removing dead servers
- Promoting backup to primary when needed
- The ping graph is confusing, just take 2 missing ping
- Pay attention to difference of initial and startup view
- Real world counterpart – orchestrator or resource provider



What you need (2)

Communication between primary and backup

- No duplicate or out of order commands (C - consistency)
- At least one server is always up (A – availability)
- Workaround network partition (P – partition tolerance)
- Reasonable response time (latency)
- Reasonable number of messages



Advice

Read the instructions carefully

Don't assume

Read the tests to understand the goal

Carefully change states

Debugging

- Print critical info
- Test predicates between state changes with assert



Tips

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